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Electrocatalytic CO₂ Reduction to Ethylene: A Carbon-Neutral Technology for a Greener and More Profitable Future

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Abstract

ADNOC's vision to achieve carbon neutrality aligns with the carbon-neutral electrocatalytic CO₂-to-ethylene (ECO₂C₂H₄) technology. In collaboration with the American University of Sharjah (AUS), ADNOC is considering converting captured CO₂ into high-value ethylene using renewable electricity. Ethylene is a crucial chemical for ADNOC's downstream petrochemical processes. After validating the ECO₂C₂H₄ technology at a 100-ton/year pilot scale, ADNOC plans to launch a joint commercialization initiative with AUS and its partner universities to deploy a 200-ton/year demonstration plant at its facilities. This project is aimed at converting CO₂ liabilities into profitable assets, accelerating ADNOC's decarbonization goals and establishing the company as a pioneer in sustainable carbon utilization.

Introduction

As global energy demand surges and the appetite for petrochemicals intensifies, the escalating consumption of fossil fuels drives a sharp rise in anthropogenic carbon dioxide (CO₂) emissions – further accelerating the pace of global warming and deepening the climate crisis. CO₂, the primary greenhouse gas driving global warming, is the major byproduct of fossil fuel combustion. Around 14% of global oil and 8% of global natural gas is used as feedstock for the petrochemical industry for making products like plastics, fertilizers, and synthetic fibers [1]. These petrochemicals are major drivers of fossil fuel demand and represent a significant non-combustion source of carbon emissions. Consequently, the atmospheric CO₂ levels are now 50% higher than they were pre-industrial revolution [2], and this has already taken its toll on climate, causing extreme weather conditions, floods, oceans acidification, and shifts in weather patterns. The deepening climate crisis, coupled with rising fossils demand, underscores the urgent need to transition to net-zero and ultimately net-negative carbon footprint technologies.

This precarious situation led nations worldwide to contemplate and devise a sustainable solution to this two-front challenge. A breakthrough in this effort was the Paris Agreement of 2015 [3], which committed

196 countries to implementing measures to limit global warming to well below 2 °C, preferably to 1.5 °C above the pre-industrial levels by 2100. While all the member countries initially submitted their Nationally Determined Contributions (NDCs), many have not updated them ambitiously [4]. As concluded by the 2023 Global Stocktake at COP28 UAE, the current commitments fall short of limiting global warming to 1.5 °C [5]. In fact, scientific assessments – including those from the Intergovernmental Panel on Climate Change (IPCC) – emphasize that reaching net-zero by 2050, needs global GHG emissions to peak by 2025, and declines by 43% by 2030 [6]. To adhere to the 1.5 °C threshold, the COP28 UAE Consensus of 2023 reinforced these critical timelines, urging nations to submit stronger 2035 climate plans by 2025 and accelerate near-term actions [7]. Key global commitments included transitioning away from fossil fuels [8] and tripling renewable energy capacity by 2030 [9], underscoring the urgency for immediate decisive action.

UAE announced a national net-zero-by-2050 goal in 2021, the first Gulf nation to do so. At COP28, it positioned itself as a regional climate leader, promoting renewable energy projects (e.g., Masdar City, solar farms) and pledging \$30 billion to a global climate fund [10]. ADNOC, as a global leader in energy innovations, aims to achieve net-zero emissions for its upstream operations (Scope 1 and 2) by 2045. ADNOC's allocation of \$23 billion to promote lower-carbon solutions by investing in clean energy and decarbonization technologies is proof of its utmost dedication to achieve net-zero emissions by 2045 [11]. ADNOC is a leader in carbon capture, utilization, and storage (CCUS), with captured CO₂ from Scope 1 injected into oil fields for enhanced oil recovery (EOR). ADNOC also captures CO₂ from other industries (such as Emirates Steel) and injects it into ADNOC's oil fields. Though the captured CO₂ is dumped into oil fields, there are other solutions of permanently getting rid of this CO₂ – economically more feasible and well-aligned with ADNOC's strategy to diversify its energy portfolio.

One of the most promising strategies to utilize CO₂ is to electrocatalytically convert it into fuels and other value-added chemicals (Fig. 1). The overall reaction (Equation 1) involves the reduction of CO₂ with water on an electrically activated catalyst. When powered by renewable electricity, electrochemical CO₂ reduction (ECR) presents a sustainable and environmentally benign approach with the potential to achieve a net-negative carbon footprint. A range of products including carbon monoxide (CO), hydrocarbons, and oxygenates can be synthesized with ECR – offering a more profitable yet carbon neutral alternative to dumping CO₂ into oil wells for enhanced oil recovery (EOR). The most important ECR products according to their global market value and commercial uses are given in the following Table 1.

Ethylene, a \$200 billion industry (Table 1), is the world's most-produced organic chemical. Ethylene is almost entirely used within the petrochemical sector, serving as a feedstock petrochemical for plastics, solvents, and synthetic fibers. Over 90% of its production goes into products such as polyethylene, ethylene oxide, ethylene glycol, vinyl chloride, styrene, etc., used in packaging, construction, textiles, and automotive applications [25]. Driven by the expanding downstream applications, the global ethylene demand is anticipated to grow steadily, with forecasts projecting sustained increments through 2034 [26]. This rising demand and the traditional energy-intensive synthesis practices that involve costly hydrocarbon feedstock and demand complex byproducts separation techniques, presents a strategic opportunity for ADNOC to leverage electrocatalytic CO₂-to-ethylene (ECO₂C₂H₄) with net-zero carbon emissions. By integrating ECO₂C₂H₄, ADNOC will not only mitigate its carbon emissions but also diversify its product portfolio with net-zero emission ethylene—strategically positioning itself in a decarbonizing global market. This shift will potentially help ADNOC insulate it from future carbon pricing liabilities and enhance access to sustainable capital markets.



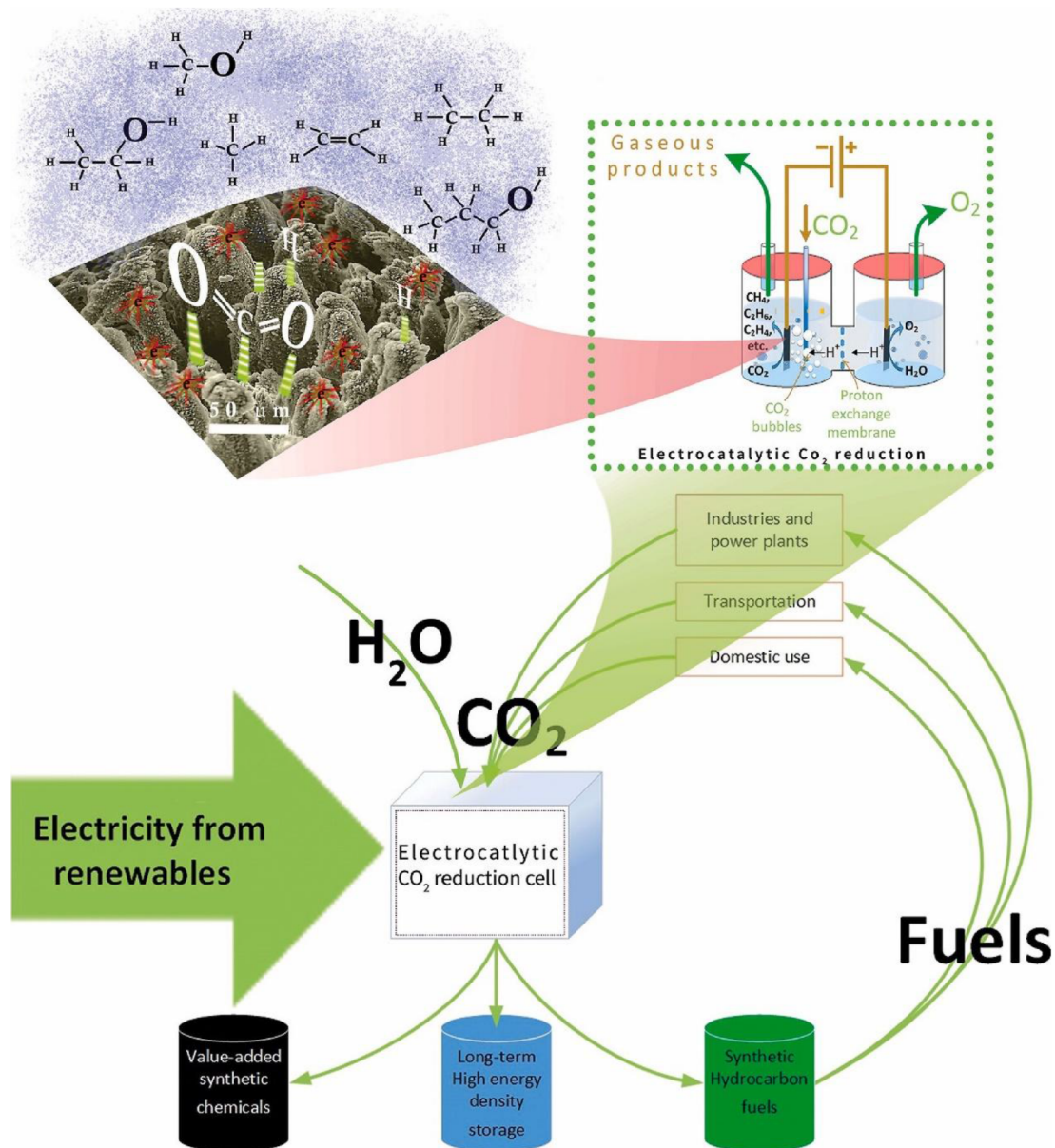


Figure 1—The concept of electrochemical reduction of anthropogenic CO₂ with water to produce fuels and other value-added products. Reproduced with permission from Materials Today Sustainability, Elsevier [12].

Table 1—Global market value of important ECR products and their commercial applications

ECR PRODUCT	GLOBAL MARKET VALUE	COMMERCIAL USES
CARBON MONOXIDE	>\$4 billion [13]	Steel manufacturing, chemical synthesis, and metal fabrication
FORMIC ACID	>\$ 1.3 billion [14]	Agriculture, animal feed preservative, leather processing, textile, cleaning products, chemical intermediate, rubber processing, pharmaceuticals, chromatography, etc.
METHANE	\$127.45 billion [15]	As natural gas for electricity generation, heating, cooking, and transportation (CNG), feedstock for methanol, ammonia (fertilizers), and hydrogen
ETHYLENE	~\$200 billion [16]	Production of polyethylene, ethylene glycol (antifreeze, PET bottles), PVC, solvent, and synthetic rubber
ETHANE	>\$13.87 billion [17]	Petrochemical feedstock for ethylene production (steam cracking); natural gas liquid, fuel blend as an energy source, anti-freeze, detergents
PROPYLENE	>\$121 billion [18]	Polypropylene (PP), acrylonitrile, propylene oxide (PO), cumene, isopropanol (IPA), oxo-alcohols, and alkylated gasoline additives
PROPYLENE OXIDE	~23.5 billion [18]	Polyether polyols, propylene glycol (PG), glycol ethers, surfactants and detergents, pharmaceuticals and cosmetics
PROPANE	\$96.07 billion [19]	LPG for heating, cooking, drying, and transportation; Feedstock for propylene and petrochemicals
ACETONE	~\$6.5 billion [18]	Solvent, methyl methacrylate (MMA), bisphenol-A (BPA), pharmaceuticals, cleaning agents, personal care
BUTENE-1	~3.2 billion [18]	LLDPE comonomer, HDPE comonomer, polybutene-1 (PB-1), plasticizers and adhesives, alkylation in refineries
BUTADIENE	~12.1 billion [18]	Styrene-butadiene rubber (SBR), polybutadiene rubber (PBR), acrylonitrile butadiene styrene (ABS), nitrile rubber (NBR), adiponitrile production, chloroprene (Neoprene)
BUTANE	>\$108.43 billion [20]	LPG, gasoline blends, feedstock for butadiene (synthetic rubber), ethylene, etc.
METHANOL	\$34.253 billion [21]	Formaldehyde (resins, adhesives), MTBE, biodiesel, and methanol-to-olefins
ETHANOL	>\$108.44 billion [22]	Fuel ethanol (E10/E85 blends), beverages, sanitizers, solvent in pharmaceuticals/cosmetics
PROPANOL	>\$3.9 billion [23]	Disinfectants, pharmaceuticals, solvents, cleaning
BUTANOL	\$10.79 billion [24]	Solvent (paints, coatings)

Technology Background & Strategic Relevance

The $\text{ECO}_2\text{C}_2\text{H}_4$ system. $\text{ECO}_2\text{C}_2\text{H}_4$ is in its early stages of commercialization, with feasibility demonstrated at lab-scale and pilot-scale levels. Ethylene Faradaic efficiency surpassing 50%, has been demonstrated for several $\text{ECO}_2\text{C}_2\text{H}_4$ reduction systems [25], [26], [27]. However, challenges remain in further improving selectivity, operation under field conditions, maintaining industrially relevant reaction rates, improving single-pass CO_2 conversion, enhancing energy efficiency, and ensuring long-term catalytic and operational stability [27], [28], [29], [30], [31].

Through research, we now know the proper conditions required to maximize ethylene yield. Among the criteria for ethylene production, the following are important considerations

- Effective adsorption of CO_2 and H_2O (or H^+) on a surface.
- The surface can provide sufficiently energetic electrons to drive multi-electron reductions.
- The surface exhibits moderate binding strength toward key reaction intermediates (e.g., $^*\text{CO}$, $^*\text{OCCO}$, $^*\text{CHO}$, $^*\text{CH}_2$).

- The surface is capable of stabilizing C–C coupling intermediates without overbinding.
- The surface can maintain a low surface coverage of water or protons, minimizing competing hydrogen evolution.

These criteria are central to a typical $\text{ECO}_2\text{C}_2\text{H}_4$ system and comprises of an ethylene selective catalyst and an electrolyzer assembly (Fig. 2) put together such that the components are complementing each other in meeting the above criteria. To overcome the inherent issues of CO_2 solubility in aqueous solutions, and to manage the $\text{CO}_2/\text{H}_2\text{O}$ ratio, flow cell electrolyzers are used. Different configurations of flow cell electrolyzers are employed to enhance the selectivity, reaction rates, and stability of the $\text{ECO}_2\text{C}_2\text{H}_4$ process [12].

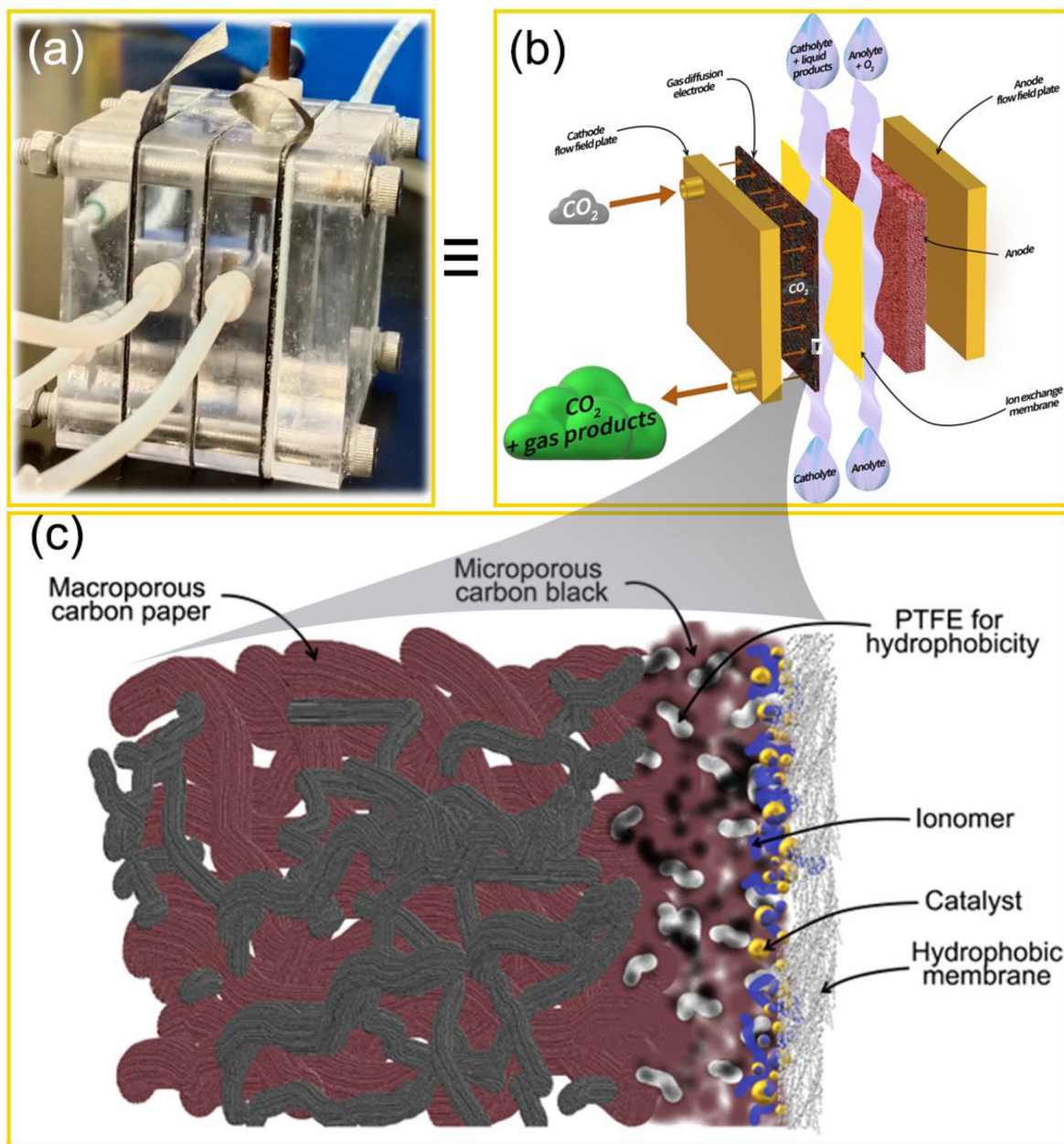


Figure 2—(a) A camera-captured image of a lab-scale flow cell electrolyzer; (b) Its schematic representation highlighting its major components, the anolyte, the catholyte, and the gas flow fields; and (c) a schematic of a gas diffusion electrode with all functional components highlighted.

ECO₂C₂H₄ as a net-zero replacement to CO₂-EOR. Repurposing captured CO₂ to produce carbon-neutral ethylene instead of enhanced oil recovery (EOR), could unlock higher revenues and carbon credits. ECO₂C₂H₄ which converts CO₂ into ethylene, not only outshines conventional ethylene synthesis in carbon-neutrality but also rivals it in cost-competitiveness – considering the costly fossil feedstocks and complex purification techniques employed to produce ethylene by conventional means.

Synergy with ADNOC's Assets. ADNOC has been pioneering CO₂ injection for EOR since 2016, particularly through its Al Reyadah project, which captures CO₂ from Emirates Steel Industries and injects it into oil reservoirs to boost recovery [32]. The company is en route to expanding its CO₂ utilization for EOR by planning a six-fold increase over the next few years [33]. Particularly, ADNOC's Habshan CCUS project, which is expected to complete by 2026, will capture and store 1.5 million tonnes of CO₂ per year, tripling ADNOC's carbon capture capacity to 2.3 mtpa [34], [35]. On the contrary, Borouge, which is ADNOC's petrochemical subsidiary, operates multiple crackers to convert ethane and propane into ethylene, propylene, and ultimately polyolefins. ADNOC, as an energy producer committed to carbon-neutrality, with petrochemical subsidiaries and established CCUS infrastructure, is strategically well-positioned to integrate ECO₂C₂H₄ into its operations. Commercial-scale CO₂ conversion into on-demand ethylene will position ADNOC to reduce reliance on fossil-based ethylene, monetize CO₂, and achieve Scope 1/3 emission cuts.

Experimental Considerations

To make the ECO₂C₂H₄ process industrially feasible, optimization of the following components is necessary.

Electrolyzer Design. Using a flow cell architecture replenishes the reactants, removes the products, maintains temperature and ionic conductivity, and minimizes pH gradients. Reducing the anode to cathode distance is important realizing high reaction currents. Higher reaction currents produce more local OH⁻ along the cathode, which is considered important for realizing high ethylene yields [27]. Microfluidic and membrane electrode assembly (MEA) electrolyzers are common and are decided based on the electrolyte, membrane, required current, and energy efficiency, etc. [12].

Ion-exchange membrane. Ion-exchange membranes are meant for allowing desired ion(s) to pass through, while blocking/suppressing the transport of others. They are also intended for reducing/blocking reactant/product cross-over. Anion and cation-exchange membranes are common, with anion exchange membranes often used in ECO₂C₂H₄ [36]. Bipolar membranes and a combination of anion-exchange and a proton-exchange membranes have also been recently reported for ethylene [37].

Catalysts Development. Cu is known for C₂₊ products [27], however, for improved selectivity, its oxides [38], [39] and alloys [40], [41], [42] are preferred. Certain Cu alloys can stabilize the intermediates, e.g., *CO for enhanced C-C coupling. Further, typically alloys are more stable compared to oxides [40], [41], [42].

Choice of Electrolyte. Acidic, neutral, pure water, and basic electrolytes are known to steer selectivity and stability. Acidic electrolytes are rich in H⁺, which promote hydrogen evolution reaction, which is the least required in ECO₂C₂H₄. Whereas alkaline electrolytes are rich in OH⁻, which is known to promote ethylene production. Neutral electrolytes contain both H⁺ and OH⁻, offering a balance but often leading to moderate selectivity. Pure water, on the other hand, lacks significant ionic content, and generally results in low conductivity, low energy efficiency, and poor reaction rates. The choice of electrolyte is critical and dictates the remaining configuration of the electrolyzer [27], [31], [37], [39].

Gas Diffusion Electrode (GDE) Design. Gas diffusion electrodes provide the triple phase boundary for the electrochemical reaction. Hydrophobic GDEs are known to favor ethylene and avoid flooding. Stable ethylene FE is directly dependent on persistent hydrophobicity to avoid flooding [12].

Suitable Ionomers. Ionomers, which are used as binders, offer a range of attributes to tune the catalyst microenvironment and influence the activity and selectivity of the catalyst. Ionomers may have a hydrophilic or hydrophobic backbone and may have either acidic or basic ions for transport/exchange. Ionomers may also have special groups with CO₂ capturing affinity, thus increasing the local CO₂ concentration near the catalyst – a requirement for ECO₂C₂H₄ of dilute CO₂ in flue gases [28], [29], [30], [31].

Our ECO₂C₂H₄ Electrolyzer System

AUS has been collaborating with partners to develop selective and stable ECO₂C₂H₄ systems. A pilot-scale 100 ton/year ECO₂C₂H₄ technology has already been validated. The technology is ready for demonstration and commercially feasible. It integrates components such as:

- High-selectivity catalysts, achieving ~65% ethylene Faradaic efficiency.
- Integrated gas diffusion electrodes and alkaline membrane-based MegaWatt-Scale Stack design.
- Dynamic grid integration, matching intermittent electricity from renewables/clean energy sources.

A 200-ton/year ECO₂C₂H₄ technology demonstrator— scalable to meet larger demands — demonstrates interesting economics and is being considered by ADNOC for deployment.

Conclusion

Carbon-neutral ECO₂C₂H₄ systems support ADNOC's net-zero pledge and UN SDG 13. Such systems create circular economy by closing the carbon loop by converting waste CO₂ into chemical building blocks. These systems qualify for IRA 45V tax credits and EU innovation fund grants. Adopting such carbon-neutral/carbon-negative technologies will potentially position ADNOC as an innovator in carbontech, appealing to ESG-focused investors. It is high time for ADNOC to unlock first-mover advantage in a carbon-to-chemicals market, to set resilient margins and replicate model for hard-to-abate industries.

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